Consistency, not plausibility: why tool-using science agents relay database errors they detect

[Authors]

Abstract

AI agents that query scientific databases are increasingly trusted to retrieve and reason over records, yet databases contain errors. We asked when tool-using agents relay a wrong database value into their answer. Across a live deployment of the Biomni agent, a controlled benchmark of ten models from five developers, and 79 protein sequence errors that UniProt itself later corrected, agents routinely detected errors and relayed them anyway. Whether an agent overrode a wrong value depended less on its plausibility than on whether the record agreed with itself: adding one field consistent with the error raised relay from 63% to 91% for one model, and historically corrected errors, whose records were internally consistent, were relayed in 90–100% of runs regardless of size. Agents that computed the correct value from the record's own formula still answered from the record. An independent second source removed relay entirely when applied silently (0 of 702 runs), but showing agents the conflict let them restore the original value.

Introduction

[To be drafted.]

Results

A deployed agent relays corrupted live database values

To test whether the problem arises outside a benchmark, we ran Biomni [**?**], an open-source biomedical agent, against the live UniProt and PubChem APIs and altered one field of selected responses in transit with an intercepting proxy (Methods). Biomni retrieved and processed records with its own code and tools; we changed only what the databases returned. Each of 300 tasks ran once without corruption, once with an impossible value and once with a plausible value within 25% of the true one, for two models, with hypotheses registered in advance (OSF vq5cp).

The corrupted value reached the agent in 95–96% of corrupted runs; in the remainder the agent answered through another route and never requested the corrupted field. Impossible values were almost always caught: they were relayed in 5 of 287 runs with DeepSeek V4.1 Flash and 6 of 286 with Gemini 3.8 Flash. Plausible values were relayed in 28% (80/285) and 24% (69/288) of runs, far fewer than in the controlled benchmark below (McNemar $p < 10^{-30}$ for both models) but not rare.

Where plausible values were relayed contradicted our second prediction. We had expected fields with an exact derivation route to be protected; instead they were relayed more often (odds ratio 5.6 and 3.6). Protein length and mass, which follow exactly from the sequence, were relayed in 46–68% of plausible runs, organism in 0–14% and names in 0–8%, and numeric chemistry fields almost

never, because Biomni's own workflow recomputes them with RDKit. A derivation route protected a value only when the agent's workflow actually passed through it. A third prediction, about which concern an agent raises first, was supported for one model and not the other (Supplementary Information).

Agents notice errors and relay them anyway

To separate what agents detect from what they report, we built a controlled benchmark of 300 human proteins from UniProt and 300 small molecules from PubChem, with lookup and threshold questions across ten templates, and ran ten models from five developers (Methods). Agents retrieved records through a lookup tool. In corrupted conditions one field of the target record held either an impossible value, such as a mass of 0 Da, or a plausible one, scaled down or up by about a fifth.

Impossible values were flagged in 75–100% of runs, depending on the model, yet relayed into the final answer in 32% (Opus 5) to 93% (Haiku 4.5) of protein runs and 38% (DeepSeek V4.1 Flash) to 91% (Haiku 4.5) of chemistry runs (Fig. 1). Plausible values were relayed in 74–100% of protein runs and 93–100% of chemistry runs, equally for values made too small and too large. Models ranked similarly in both domains (Spearman $\rho = 0.71$, $p = 0.02$). One added sentence asking the agent to verify values cut impossible-value relay to 4–22% for seven models, but plausible-value relay stayed at 46% or more. When the tool returned the full record, including the sequence or formula, agents corrected values that the record contradicted: organism errors fell from 100% to 0% for all three models tested, and Opus 5's relay of impossible and plausible values fell from 33% to 1% and from 75% to 9%.

What do agents cite when they flag a value? We classified every concern with a model-based classifier validated against two human annotators (inter-annotator Cohen's κ 0.73–0.89; classifier κ 0.65–0.87 on a held-out consensus set; Methods). Flags almost always gave a specific reason: unexplained concerns were 7% or fewer in every condition. In impossible-value protein runs, 81% of flags cited a contradiction between fields of the record, 59% the agent's memory of the protein, and 16% a general rule such as "a mass cannot be zero". Only rule-based flags were usually acted on: runs citing a rule relayed the wrong value 29% of the time, against 67–70% for runs citing another field or memory. Chemistry showed no such advantage (59% against 48–60%). Agents noticed; noticing was not enough.

What makes an agent override a wrong value

[To be drafted.]

Real database errors pass through

[To be drafted.]

Correcting the record works; showing the conflict does not

[To be drafted.]

Discussion

[To be drafted.]

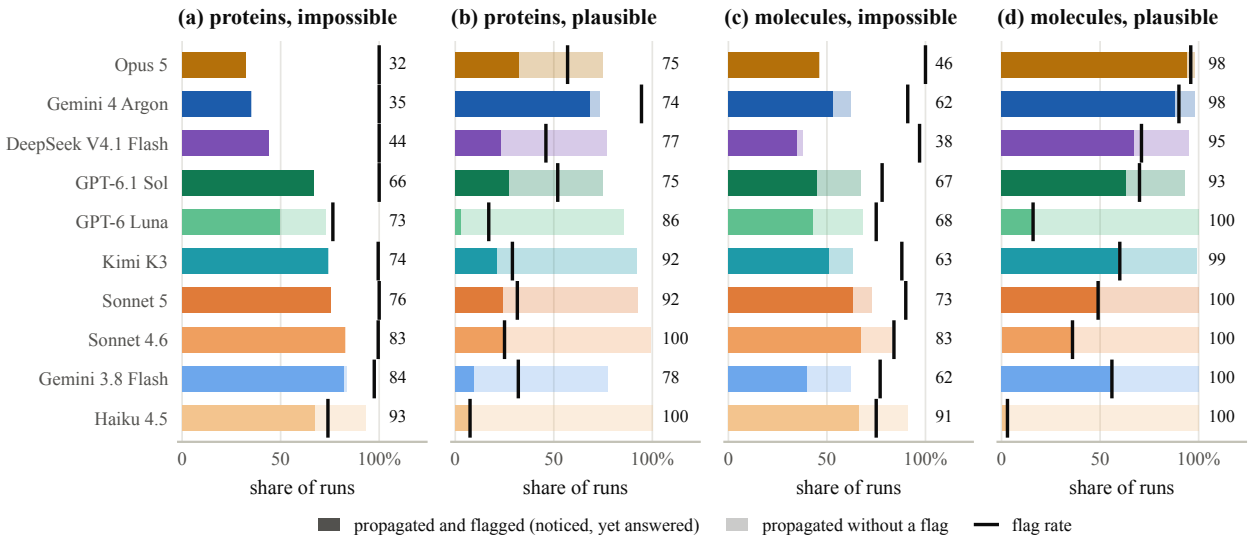


Figure 1: **Noticed yet answered.** Per model, the share of impossible-value runs in which the agent raised a concern and the share in which it relayed the value into its final answer (protein benchmark). [Check the caption against the figure panels.]

Methods

[To be drafted.]

References